

Development of On-Board Diagnostics for Car and it's Integration with Android Mobile

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Abstract— On-Board Diagnostics II (OBD-II) port began as a means of extracting diagnostic information and supporting the right to repair. Self-driving vehicles and cellular dongles plugged into the OBD-II port were not anticipated. Researchers have shown that the cellular modem on an OBD-II dongle may be hacked, allowing the attacker to tamper with the vehicle brakes. Advanced driver-assistance systems(ADAS), self-driving features and other vehicle functions may be vulnerable as well. The industry must balance the interests of multiple stakeholders including Original Equipment Manufacturers (OEMs) who are required to provide OBD function, repair shops which have a legitimate need to access the OBD functions, dongle providers and drivers. OEMs need the ability to protect drivers and manage liability by limiting how a device or software application may modify the operation of a vehicle.

The paper consists of two components: Linux running Texas Instruments(TI) OBD-II hardware and OBD-II application running on Android mobile. Design is explained in detail for both. Wireless communication(Bluetooth) is used between Android OBD-II application and OBD-II hardware. The TI hardware is main component which is responsible for communication to cars ECU (Electronic control unit) and as well as mobile. Work involved is development of various software component on TI hardware and android Mobile application.

With Machine learning, artificial intelligence is becoming integral part of the futuristic automotive evolution. This paper can act as a benchmark for users to manage their automobiles easier.

Keywords— Diagnostics, Cellular, OEMs, Wireless communication

I. Introduction

Computer programs and embedded systems have been part of the cars from long time. Each car running on the road has tons of data stored and available for service stations, car manufactures and second-hand car sellers. Most of the car owners are not aware the price of the second-hand car is decided based on the information available on these computers in the car. With the advent of smartphones, accessing the data from car computers using wired and wireless technology for car users has become very easy. The information available in cars can be seen by the owner with the help of their

smartphone and cloud technologies. As per the recent changes in the automobile industry paradigm, automobiles are mandatory to develop on eco-friendliness, safety, and comfort. Automobiles were considered merely means of transportation with independent hardware. As many additional services and safe and convenient automatic internal control systems have been introduced to provide more comfort and safety of the customers', however, IT technology must be added as a software for providing electronic systems in the powertrain, body, chassis, and infotainment systems, via the vehicle network (control area network(CAN), local interconnect network(LIN), FlexRay, MOST, etc.), to ensure eco-friendly and high-quality customized services. Most smartphones have a Bluetooth device for short-range wireless communication as well as a 2G/3G/4G module, and diverse products are being developed using this feature. Most automobile multimedia devices.

We have the second-generation on-board diagnostics (OBD-II) automobile diagnosis standard that was devised for diagnostics and emission control. This provides an opportunity to develop new software's that provides more information to car owners with diagnosis services. Improved functions for automobile maintenance and diagnosis, and more convenient device control can be provided. In this paper, a vehicle diagnosis is developed, in which vehicle information is retrieved from the Electronic Control Unit(ECU) according to the OBD-II standard, to maintain and diagnose the vehicle using the Bluetooth wireless network technology, along with Android smart phone. With this program, the diagnostic data can be precisely

checked in real time to maintain the optimal vehicle conditions. This technology has also been applied to the reduction of Carbon dioxide (CO₂) emission and can improve vehicle's eco-friendliness.

There are two main occurrences that will cause the average car owner to panic. The first one, the appearance of the check engine light on the dashboard, often leads to the second one, the appearance of a large bill at the car service station. For the most car owners, lights on the dashboard are a mystery to most people, and can generate considerable anxiety. The doubt would come to car owners mind, what is the issue with car and what changes has happened with the car to prompt this seemingly-foreboding message? And since the light invariably turns on when funds in your bank account are at their lowest, as if it somehow knows, it's not uncommon to ignore it for months and maybe even years. If the vehicle seems to be running fine, why rush to investigate? Investigating, after all, is often expensive. When the check engine light comes on in your car, that means the car's computer has encountered an error of some kind. This could be anything from a loose fuel cap to a dead oxygen sensor. That error is logged as a code, and remains in the computer's memory until it is cleared, either with a special tool or a predetermined reset of so many engines starts without the condition recurring. The system in a car responsible for generating these code is called on-board diagnostics, or OBD, and as of 1996, the OBDII protocol was made mandatory for all cars to be sold in the United States. In Europe, they have EOBD(Europe OBD), in Japan JOBD(Japan OBD), ADR(Australian Design Rules) in Australia and BHARAT Standards in India. These codes are checked with what's called a *scan tool*, which is a standard piece of automotive technician hardware. They typically range from the relatively simple to rather more involved, but the premise is the same: a small piece of hardware with a spartan Liquid crystal display(LCD) screen, a few buttons, and a primitive operating system. They are pre-loaded with the trouble codes for a variety of manufacturers and help quickly assess engine performance. The trouble is that bringing your car to the service centre. Not only do you have to make time for this, but it also ensures that you'll end up

hearing that your car needs work and that a large bill is forthcoming. Aside from that, unless you're working with a shop you really trust, you never quite know if what they're telling you is totally true, or the urgency of the issue. While it's true that you can go to some auto parts stores to get the code read and even cleared, there could be some pressure to buy the parts right then. That's why they do it, after all. It's a good service, but they need to sell parts at the end of the day.

Application on Android smartphone can show different vehicle parameters being displayed in from BT(Bluetooth) enabled OBD device.

The paper consists of following sections: Section II explains about related work; Section III describes about system analysis; Section IV deals with system design and section V discusses about results. Finally, the paper includes conclusion and future scope.

II. Related Work

USA government has introduced federal law for OBD to all vehicle manufactures to meet the OBD standards. In 1996, the second generation OBD was introduced. To meet On-board diagnostics second generation standard, all the car manufactures must provide on board computer. This On-board computer has to perform the monitoring activity and carryout emission test on vehicle. To make sure the vehicle is pass the acceptable level of emission test. The emission test level is defined by each country, so very country has different standard for emission. The second generation of ODB has standard hardware, with 16 pin connectors. OBD-II eliminates the manufactures specification, this makes all the car manufacture need to follow the same hardware on the car.

Sang Hyun et al. [1] has described about development of OBD via Control Area Network (CAN) for HVI (Human Vehicle Interface) technology. Antoni Jankowski et al. [7] has shown some aspects of OBD at Poland. They have measured the quantitative efficiency of the system OBD in characteristic operation conditions, securing of 90% a probability. C. Varun et al. [2] have explained about the Ethernet based OBD-II, where the driver can find out the faults in car on the

spot and this does not require an OBD scan tool for it. If in case, the vehicle got struck in remote areas, proposed OBD system inside the vehicle can be connected to the vehicle service centre by providing internet facility.

III. System Analysis

Car with on-board computer controls the breaks, transmission, traction, engine and other cars components. The malfunction indicator lamp is located on the instrument panel on the car's dash board. The color of the lamp/light either yellow or red in color. The malfunction indicator generally off, it will glow the lamp/light based on the car's component failure. Emission level crosses the 1.5 times prescribed by the governments emission level. The initial data stored in the read only memory of the on-board computer is called Frame data freeze or Freeze Frame Data. Once this Frame data freeze is written, this data cannot be modified by the any user apart from the manufacture. Manufacturer can modify later this ROM but government can be verify, how many time this ROM is modified along with date and time. The main components considered for OBD are: Car , Customized Bluetooth based OBD II reader, Internet connection, Android smartphone with Bluetooth connectivity, Car driver and Cloud system as shown in Fig. 1.

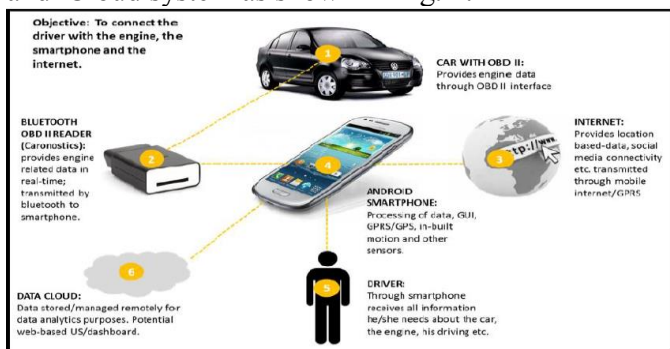


Fig. 1: System Overview: Main Components

Cars having electronics control unit(ECU) provides lot of data that could help car drivers to save lot of issues in the car to help drivers and mechanics before start analyzing the problem. Before giving to service centre user can note down the ECU data and verify the data once the service done, this avoided the extra charges from service centre. *OBD port* is the entry point into the car provides lot of information that would enable driver and mechanics. TI processor based OBD connector can

be connected to 16 pin connector, with adding Bluetooth (BT) for communication. Connecting the car to internet enables real time tracking of issues and information. Present system does not support *Internet connection* directly because OBD II hardware does not have Wi-Fi or Ethernet inbuilt. The data can be sent to cloud system using mobiles internet connection. The mobile will get the data from the OBD II hardware using BT communication and later mobile send these data to cloud using its modem to cloud. The cloud client will run on the mobile which will send data to cloud server. The cloud server can be hosted at any place in the world. OBD II application is developed and integrated with *Android mobile*. Android mobile will get the data from the car, using the BT communication with OBD II hardware. Android phone process this data obtained on the online before uploading the data into cloud. *Human intervention* is very important to understand the issue and enable to provide relevant information to be feed. The driver or end user of car can get the complete car's information. This information gives user fair idea on, is any fault in car. If car has fault user can go for service centre. *Cloud* takes service from IT infrastructure provider on software as a service. The data availability on cloud make end user can login to his/her personal account to the cloud server and check information about car. This can be used as tracking system for car.

The main hardware and software requirements considered are: OBD -II hardware from TI, Serial debug terminal, JATG, USB to serial cable, 2GB MMC, SD card reader, Cloud Client and Server, Latest android mobile, U-boot source code, Android above ginger bread OS, Eclipse latest version, ARM Cross compilers for application and Linux kernel, CAN, BT, Wi-Fi, Touch, Display. Gyro sensor device drivers, Trace32, Git repository, JAVA compiler, Python and shell scripting, JSON parser and Doxygen.

IV. System Design

System design deals with the algorithms for Mobile application and OBD-II hardware.

Mainly three modules are considered in the design namely: *Init Module* : To run OBD II application it

has to first register with touch screen driver which helps the touch events on the mobile for specifically with OBD II application. To start this application user has to touch the OBDII icon on the mobile. The gyro sensor which helps the screen rotation. Get the screen orientation to display the OBD II application within the screen area. Before displaying application contain on the screen need have screen information. This will help to display on properly on any display size of mobile. Once you get the screen information, you can start manipulating and drawing the icons on the mobile screen. Based on the screen you can adjust the size of the icon dynamically. If icons are more and not able fit in the single view on the mobile screen than need to give scrolling option to display. This completes the UI design for OBD II application. To connect mobile with OBD II hardware, BT is required. To access BT hardware present on the mobile, first application need register with android BT middleware software stack. The android BT middleware is register with Linux BT device driver. The BT device driver is responsible for accessing the Bluetooth hardware. The application checks first the Bluetooth ON or OFF. If BT is OFF than application makes BT ON. The application writes "TRUE" into Bluetooth specific file in the sys interface in the root file system. Once the BT on the paring mechanism starts, the mobile will start scanning the Bluetooth device in the BT range. If the OBD II hardware is running it will show in the mobile. On mobile BT select the OBD II hardware as pair. The application asks the password, enter password it will get paired with mobile and OBD II hardware; *Communication Module*: responsible for data transfer between the android mobile and OBD II hardware. The communication is based on the eventing mechanism. This module is wait for new data from the OBD II hardware. Once the data is received from OBD II hardware, data will display on the related tab on the android OBD II application; and *Exit Module*: responsible for BT unpairing. Application will unregister with BT hardware. Resource allocated for BT is freed when unregistering with BT. Unregister with touch screen and gyro sensor, which free the resource acquired by OBD-II application when it was registered with touch screen and gyro in init module.

Complete OBD-II system for Android application is indicated in Fig. 2.

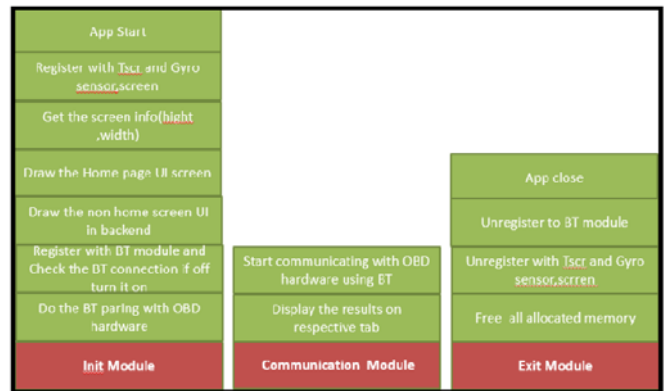


Fig. 2: OBD-II Android application design

Algorithm-1: Mobile application:

Step1: Initialize the Bluetooth
 Step2: Wait for Bluetooth connection
 Step3: Get the supported commands
 Step4: Send the commands to OBD-II hardware
 Step5: Wait for OBD-II hardware response
 Step6: Display the result on the mobile display
 Step7: Exiting from application (any Error this step has to call).

Algorithm-2: OBD-II hardware :

Step1: Initialize the OBD-II hardware
 Step2: Initialize the Bluetooth hardware
 Step3: Wait for Bluetooth connection.
 Step4: Send supported command to Bluetooth client.
 Step5: Wait for Bluetooth command
 Step6: Forward the command to ECU.
 Step7: Response from ECU forward to Bluetooth client.

Data flow diagram for OBD-II is shown in Fig. 3.

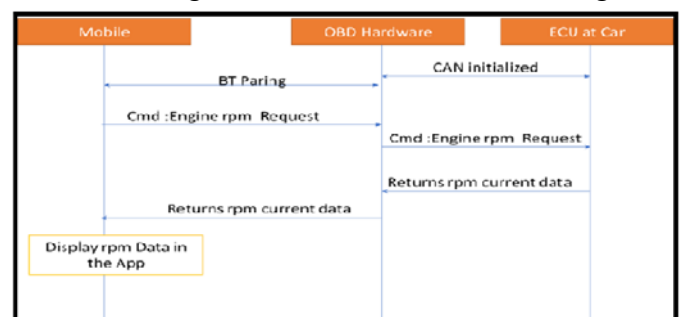


Fig. 3: Data flow in OBD II system

V. Results and Discussion

The OBD-II hardware is connected to a Skoda car. OBD-II hardware gets data related to Air Intake Temperature and send it to android mobile smart application. This procedure will continue with other parameters like Ambient air temperature, Engine coolant temperature, Barometric pressure, Fuel pressure, Intake manifold pressure, Timing advance, DTC number, OBD standard, PID check, Equiv ratio, vin as shown in Fig. 4.

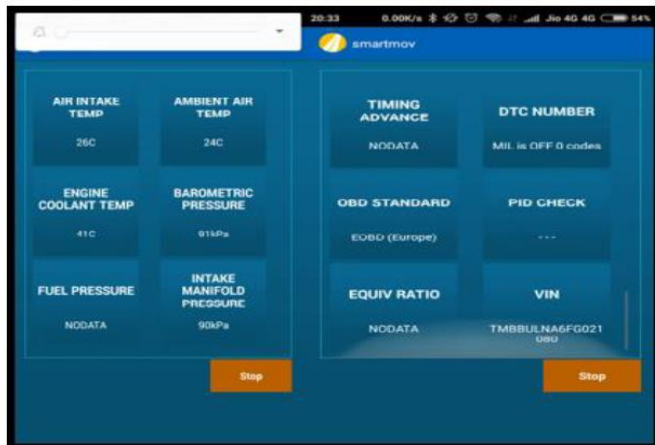


Fig.4: Data obtained after OBD connected to a car

Fig. 5 shows fuel type displays the type of petrol, diesel, gas electrical type. The Fuel consumption displays the consumption of the fuel. Similar displays the other fuel related. Engine related data like load, revolutions per minute (RPM) will display. Other parameter like distance covered, speed will get displayed.

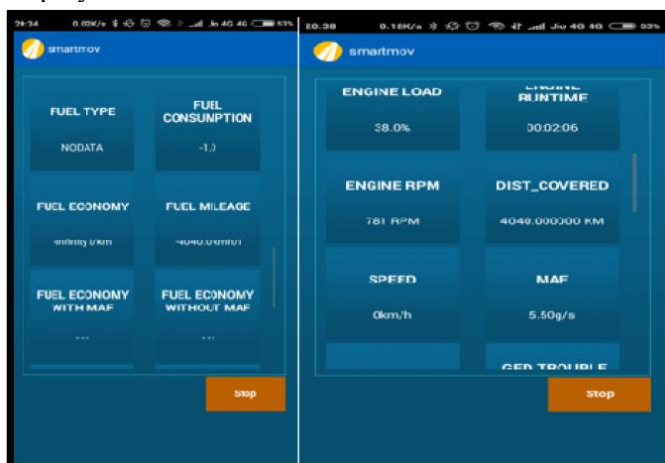


Fig.5

Fig. 6 gives information's of throttle, trouble code, fuel economic without MFA, Fuel level, fuel type, fuel consumption, and various data related fuel parameter. Fig. 7 indicates the timing advance, DTC(Deutsche Tourenwagen Challenge) number PID(parameter ID), OBD standard, and equity ratio.

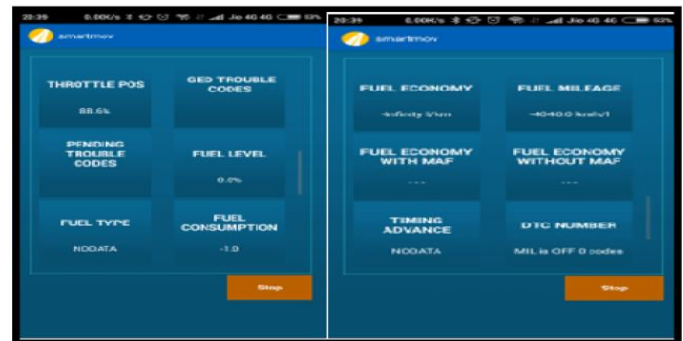


Fig.6

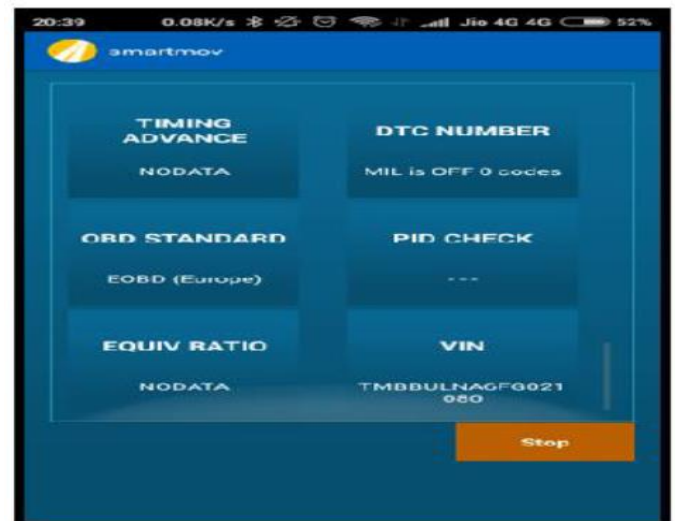


Fig.7

The data stored on cloud, which is sent by the android mobile is shown in Fig.8.



Fig.8

Conclusion and Future Scope

With Machine learning, artificial intelligence becoming integral part of the futuristic automotive evolution this paper would be an entry point for users to manage their automobiles easier. OBD-II port based diagnostics solution will provide proper data to user/driver to take smarter decision in handling for emission control, reducing the risk of untimely vehicle breakdown. It will also be very useful in Electric cars and driver assistance systems. With global warming becoming a major issue, maintaining a healthy automobile has been mandated by the governments across the globe. The hardware that will have firmware and software stack that will manage and notify the details to smart phone application that would interact with the automobile via secure short range wireless protocols such as Bluetooth/Wi-Fi. Smartphone application will help the driver in understanding the issues that vehicle can produce and keep the logs of the important incidents into a cloud storage that can be used for better maintained of the automobile and getting the higher valuation.

The present system has only BT interface with OBD-II hardware. Having only BT communication with external world, this makes limiting the BT range communication between the two BT device. Other limitation is – presently the screen drawing on the mobile phone is statically configured. If the car has more parameter than those parameters won't be displayed or if car has less parameter than it will display the data is not available.

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